Task-Dependent Neural Geometry Is Layer-Specific: An Exploratory Within-Subject Analysis of ds004511

Chart-Jacobian, one-step, and reachability estimates are distinct objects

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Abstract We asked whether resting state, cognitive control (CC), and gambling (GG) differ in explicitly non-interchangeable summaries of a subject-anchored neural chart. We reanalysed 134 EEG recordings from 45 OpenNeuro participants (45 Rest, 45 CC, 44 GG; one documented GG recording absent). Within-subject Wilcoxon tests and BH-FDR were applied in separate endpoint families. GG minus Rest increased the local chart-Jacobian spectral abscissa (median difference 0.01248, paired $d_z = 1.42$, $n = 44$, $q = 5.61 \times 10^{-}10$); the direction persisted when every task was re-aggregated over the same first 400 s ($d_z = 1.12$, $q = 2.76 \times 10^{-}8$). The independently identified lag-1 one-step spectral abscissa did not increase ($d_z = -0.33$, $n = 32$, $q = 0.18$). In the identified subset, GG reduced L2 rotation ($d_z = -1.10$, $q = 8.04 \times 10^{-}6$) and L3 generator rotation ($d_z = -1.08$, $q = 8.04 \times 10^{-}6$). Reachability amplitude summaries were available for 131/134 recordings; all three missing certificates failed the same numerical PSD gate, one in each task. Because task durations did not overlap and reachability was exported only as recording-level scalars, its large contrasts are not established as duration-independent. The effective-dimension reachability readout was degenerate and is not interpreted. These export-level results establish a dissociation among estimators, not a neural mechanism, clinical biomarker, or confirmatory task effect.

Author Summary

The same EEG data can support different mathematical summaries of change. Gambling altered a local chart-Jacobian quantity but not its separately estimated one-step analogue. Occupancy, local flow, one-step dynamics, and reachability must therefore be tested and reported as different questions.

1. Introduction

Neural-state summaries are often discussed as though their names imply a common object. They do not. Occupancy describes where a recording lies in a declared chart; a local chart Jacobian describes a local derivative of that chart; an identified one-step map describes a fitted transition relation conditional on support; and a reachability summary describes a finite-horizon construction. Agreement between them is an empirical result, not an algebraic entitlement. This estimator separation follows the measurement-boundary formulation of Noetic Diffusion Theory (NDT), in which MNPS is one operational chart and no available chart quantity is automatically a latent neural law (Langell, 2026).

Core question

In a fixed, subject-anchored export, do task contrasts agree across local chart-Jacobian, identified one-step L2/L3, and reachability summaries, or do they expose estimator-specific structure and limitations?

2. Methods

2.1. Dataset and fixed measurement contract

The frozen export contains 134 recordings from 45 healthy participants: 45 Rest, 45 CC, and 44 GG recordings from OpenNeuro ds004511 (Makowski et al., 2026). One participant has no recorded GG EEG; this is retained as documented missingness rather than imputed. The input contract is the canonical MNPS `[m,d,e]` chart and

its stratified 9D definition, with coordinates declared `subject_anchored`. No analysis retunes a measurement against task differences.

2.2. Endpoint families and statistics

The first family contains chart occupancy, local chart-Jacobian invariants, L1 local-law quantities, and recording-level reachability. Its substantive correction contains 45 pairwise tests; rank-degenerate `reach_d_eff` is retained as QC but excluded from BH correction and significance sorting. The second, separate 36-test family contains chart-Jacobian alpha as a bridge endpoint alongside lag-1 one-step L2/L3 quantities. Unless explicitly labelled otherwise, chart-alpha q values quoted in the main text come from this bridge family, avoiding a choice between duplicate corrections after seeing the result. Chart-Jacobian alpha is the spectral abscissa of a local chart Jacobian; L3 alpha is the spectral abscissa of the matrix-log generator proxy `log(Phi)/dt` for an identified direct lag-1 map (4 s), not the spectral abscissa of `Phi` itself and not an Ito generator. They are not two labels for one estimator. Lag-2 was attempted, but only 13 direct L2 maps and 10 L3 proxies were computed (8 s; task-imbalanced), too sparse for a panel. L4 `reach_a_q` and `reach_v_norm` are amplitude summaries rather than validated capacity or dimension measures.

Recording values were reduced to subject-by-task medians. Pairwise later-minus-earlier contrasts used complete-case Wilcoxon signed-rank tests (Wilcoxon, 1945), paired d_z, and BH-FDR (Benjamini & Hochberg, 1995) within the relevant panel. The L2/L3 pairwise panel's reported q is the global 36-test Wilcoxon family; its Friedman table is a separate 12-test family. Friedman tests use complete triplets. Bootstrap summaries use 10,000 draws with seed 20260921. This is an exploratory discovery analysis on one export: `exploratory_discovery_20260921_g4_not_locked`.

To test the estimator contrast directly without comparing raw endpoint units, we also ran a separate exploratory interaction analysis on the 121 recordings with both production-supported alpha estimators. Each estimator was centered by its pooled common-support recording median and scaled by 1.4826 times MAD before task contrasts were formed. The interaction for each complete subject was `(chart-alpha_z later - chart-alpha_z earlier) - (L3-alpha_z later - L3-alpha_z earlier)`. Wilcoxon signed-rank tests and BH-FDR over the three task contrasts form a separate three-test family; 100,000 sign-flips and 10,000 bootstrap draws are robustness summaries. This estimand is normalization-dependent and exploratory, not a raw-unit equality test.

Because recording duration differs strongly by task, a bounded post-discovery sensitivity diagnostic re-aggregated the already exported chart-Jacobian alpha series over the deterministic first 100 four-second windows (400 s) of every recording. It retained the production finite-value rule (at least 30 finite windows and at least 50% valid), introduced no padding or re-fitting, and corrected the three fixed-prefix alpha contrasts as their own sensitivity family. L4 could not be re-estimated on this prefix because the export contains recording-level certificate scalars rather than per-window L4 series; duration associations for L4 are therefore audit-only.

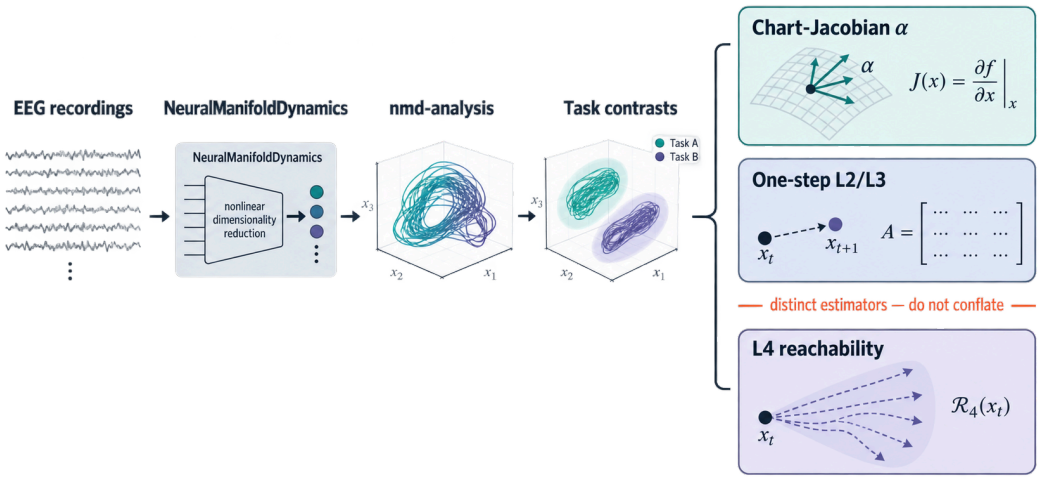


Figure 1: Conceptual analysis map. EEG recordings are processed through NeuralManifoldDynamics and nmd-analysis before task contrasts are formed. Chart-Jacobian, identified one-step L2/L3, and L4 reachability are separate estimator classes. This illustration contains no empirical values or biological interpretation.

3. Results

3.1. R1: Coverage and quality control

All 134 recordings had a finite included chart-Jacobian alpha. Lag-1 one-step quantities were identified for 121/134 recordings (115/134 for the history-conditioned L2 quantity); hence the L2/L3 paired samples are smaller than the chart-Jacobian samples. Three recordings lacked finite included `reach_a_q`: one Rest, one CC, and one GG recording from different people. Each was marked invalid after `post_dtype_psd_failure` at gate F. This is a numerical certificate failure pattern, not evidence for task-specific absence.

Object	Finite recordings	Interpretive status
Chart-Jacobian alpha	134/134	Included; local-chart quantity
Lag-1 one-step L2/L3	121/134	Included only on identified support
History-conditioned L2	115/134	Secondary identified-support quantity
L4 amplitude summaries	131/134	Included with invalid certificates visible
L4 effective dimension	131/134	Method-limited; not tested as dimension

Table 1: Endpoint coverage. Missingness changes the eligible paired sample; it is not replaced by zeros.

3.2. R2: Chart-Jacobian and one-step alpha dissociate

GG minus Rest increased chart-Jacobian alpha (median difference 0.01248, $d_z = 1.42$, $n = 44$, $p = 1.56 \times 10^{-}11$, $q = 5.61 \times 10^{-}10$). The identified one-step L3 generator-proxy spectral abscissa did not show a positive or FDR-significant counterpart ($d_z = -0.328$, $n = 32$, $p = 0.129$, global-36-test $q = 0.178$). These are different complete-case subpopulations, not a formal test of a difference between operator effects; on the same 32 GG–Rest pairs, the chart-alpha contrast remains positive ($d_z = 1.461$, median difference 0.01271, $p = 3.26 \times 10^{-}9$). CC minus Rest likewise increased chart-Jacobian alpha ($d_z = 0.433$, $n = 45$, $q = 0.0167$), while the L3 proxy was uninformative on its smaller support ($d_z = -0.103$, $n = 32$, $q = 0.595$).

The fixed-prefix sensitivity retained all 134 recordings and the direction of every chart-alpha contrast. GG minus Rest remained positive (median difference 0.01450, $d_z = 1.119$, $n = 44$, $p = 9.20 \times 10^{-}9$, three-contrast $q = 2.76 \times 10^{-}8$); CC minus Rest was 0.00593 ($d_z = 0.394$, $n = 45$, $q = 0.0124$), and GG minus CC was 0.00995 ($d_z = 0.648$, $n = 44$, $q = 7.21 \times 10^{-}5$). This supports robustness of the chart-alpha direction to equalized sampling support, not equivalence of full-recording and fixed-prefix estimands.

The separate standardized interaction test found a positive chart-versus-L3 difference for GG minus Rest (median interaction 1.888, $d_z = 0.966$, $n = 32$, Wilcoxon $p = 6.79 \times 10^{-}6$, three-test $q = 2.04 \times 10^{-}5$; sign-flip $p = 1.0 \times 10^{-}5$) and GG minus CC (median 1.229, $d_z = 0.771$, $n = 43$, $q = 2.13 \times 10^{-}5$). CC minus Rest was smaller and not FDR-significant (median 0.555, $d_z = 0.327$, $n = 32$, $q = 0.0837$). This is a formal exploratory test of the declared standardized interaction estimand on common support; it does not make the raw chart and L3 units commensurate or establish estimator equivalence.

Primary dissociation

The GG–Rest chart-Jacobian alpha result is not a one-step L3 result. The raw endpoint rows use distinct estimators and supports; a separate outcome-blind standardized interaction test on 32 common subjects was positive ($q = 2.04 \times 10^{-}5$). That interaction is normalization-dependent and exploratory, not a raw-unit effect-difference or estimator-equivalence test.

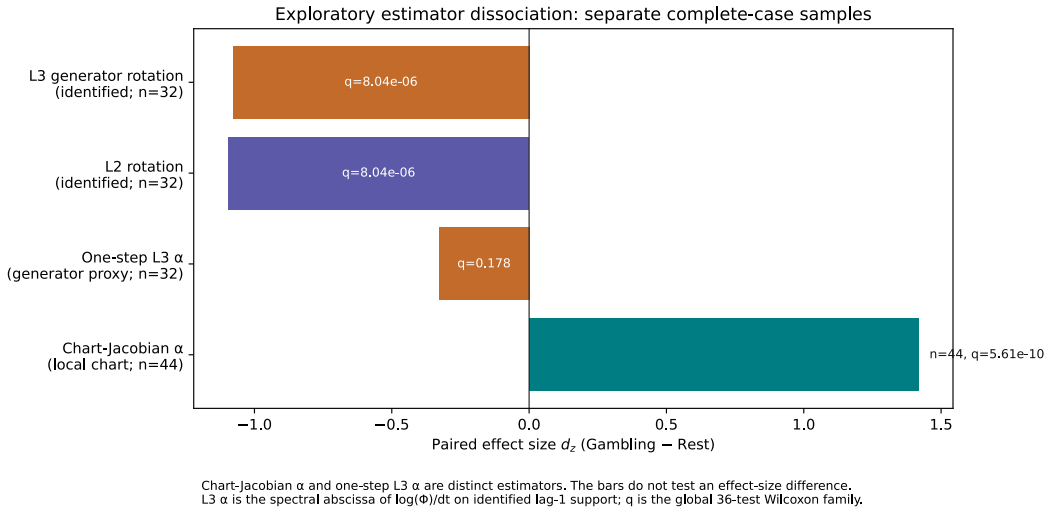


Figure 2: Exploratory GG–Rest estimator summary derived from the frozen result tables. The displayed complete-case n and global 36-test Wilcoxon BH q are estimator-specific. The bars describe distinct objects and do not test a difference between their effect sizes.

3.3. R3: Identified-support rotation results

On the 32 complete GG–Rest one-step pairs, GG decreased L2 `operator_rotation_rate` ($d_z = -1.10$, median difference -0.03589, $q = 8.04 \times 10^{-}6$) and L3 `one_step_generator_rotation_norm` ($d_z = -1.08$, median difference -0.05222, $q = 8.04 \times 10^{-}6$). These results are internally coherent within the identified subset, but they do not generalize automatically to the 13 recordings without lag-1 support.

GG minus Rest metric	n	d_z	p	**BH q**
Chart-Jacobian alpha	44	1.42	1.56e-11	5.61e-10
One-step L3 spectral abscissa	32	−0.328	0.129	0.178
L2 rotation rate	32	−1.10	6.91e-7	8.04e-6
L3 generator rotation norm	32	−1.08	6.91e-7	8.04e-6

Table 2: Key exploratory GG–Rest contrasts. Jacobian alpha and L2/L3 rows belong to distinct endpoint families.

3.4. R4: Reachability QC boundary

Finite L4 amplitude summaries differ strongly in the exploratory panel, but their interpretation is bounded by the certificate construction, the three invalid records, and recording duration. Task-duration ranges do not overlap (Rest 480–596 s; CC 904–1324 s; GG 2396–4652 s), and within GG both L4 amplitude scalars correlate positively with duration (Spearman $\rho \approx 0.35$, uncorrected $p \approx 0.02$). Because L4 is not available per window, this audit cannot remove the duration alternative; L4 contrasts are therefore not presented as duration-robust effects. In contrast, 109 of 131 finite `reach_d_eff` values were within $10^{-}6$ of 1 (median 1; maximum 1.31377). Its small-residual Wilcoxon rows are therefore not reported as capacity or dimension results.

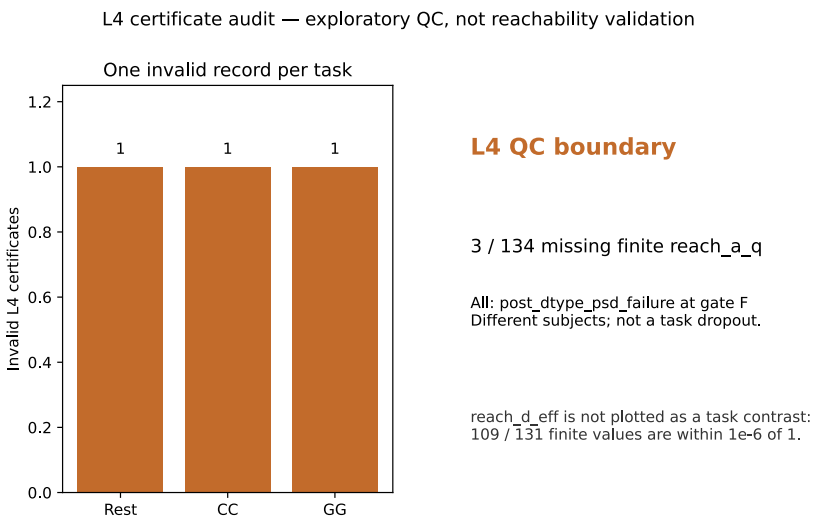


Figure 3: L4 certificate QC derived from the frozen audit. The three invalid `reach_a_q` certificates are evenly distributed across tasks; `reach_d_eff` is shown as a method limitation, not a dimensions result. This figure is a QC summary, not reachability validation.

4. Discussion

4.1. What this study establishes

On this fixed export, task contrasts are layer-specific. A strong GG–Rest chart-Jacobian alpha contrast does not entail a positive, or even significant, one-step L3 alpha contrast. The identified subset instead shows reduced rotation summaries. This is a negative constraint on claims that one scalar can stand for local flow, an estimated one-step generator, or future accessibility.

4.2. Limitations and next gate

The analysis does not identify a biological mechanism of gambling, deception, or cognitive control. It does not establish causal control, individual-level prediction, a consciousness measure, or a validated mapping from MNPS to a latent neural operator. Subject anchoring, one export, partial L2/L3 support, distinct BH families, task-duration imbalance, and the interaction's normalization choice constrain the claims. The fixed-prefix result narrows the duration concern for chart-Jacobian alpha only; it does not validate L4 or the identified one-step estimators. A bounded QC audit found that all 13 absent lag-1 maps failed the existing out-of-fold fidelity gate (`one_step_fit_not_better_than_baseline`), despite regular 4-s grids and 120–260 lag-1 pairs; they comprised 12 Rest, 1 CC, and 0 GG recordings. This diagnoses a reproducible support boundary rather than restoring a map or resolving selection bias. Independent replication on a locked export and declared multiplicity plan is required before a confirmatory task claim.

5. Acknowledgements

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Author Contributions

RL: Conceptualization, Methodology, Software, Data Curation, Analysis, Writing, and Editing.

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Data Availability Statement
OpenNeuro ds004511 is publicly available under CC0. Article-local code, pinned configuration, result tables, and provenance are in the accompanying NeuralManifoldDynamics repository.

Competing Interests
The author developed NDT, MNPS, and software used in this analysis.